Notes on forgetting

Why reviews should be spaced, and how far apart

Read a page today and by next week most of it is gone. Not all of it, and not at random: what fades first is the detail you met once and never used again. In 1885 Hermann Ebbinghaus measured this on himself with lists of nonsense syllables and found that retention falls fast at first and then flattens. The shape he drew is now called the forgetting curve.

The curve has a property that matters more than its shape. Each time a memory is retrieved, the next decline is slower. A review that comes just before you would have forgotten strengthens the memory far more than a review that comes while it is still fresh. Retrieving something that is easy to retrieve teaches you very little.

This is the spacing effect. It is one of the most replicated results in experimental psychology, and one of the least used. Students cram because cramming works for tomorrow's exam; the cost arrives a month later, quietly, when the material has to be learned again.

Stability and retrievability

Modern schedulers describe a memory with two numbers. Stability is how long the memory lasts: the number of days until the chance of recall falls to ninety per cent. Retrievability is the chance of recall right now. Stability grows with each successful review; retrievability falls with each day that passes without one.

A scheduler's job is to ask for a review when retrievability reaches a target, say ninety per cent, and no sooner. Ask too early and the review is wasted. Ask too late and the memory has to be rebuilt from scratch, which costs more than the review would have.

The intervals that fall out of this are uneven by design. A new card might come back after a day, then four, then eleven, then a month. A card you keep failing comes back tomorrow. Over a year, a well-learned card is seen perhaps six times.

Why the margin matters

Most of what we read, we read once. Highlights pile up in a sidebar nobody opens again. A highlight records that something seemed important; a question records what you would need to remember about it. Writing the question is the act that turns reading into something you can review.

Good questions are small. They ask for one fact, one step, one distinction. A card that asks for a paragraph will be graded generously and remembered poorly. A cloze, which hides a single word inside a sentence you wrote yourself, keeps the context and tests the part that matters.

The rest of this sample is filler so the reader has pages to scroll through. Odyssey keeps your position, your notes and your review history on your own server.

A worked schedule

Suppose a card has a stability of three days after its first review. On day three its retrievability has fallen to ninety per cent and it comes back. You recall it, and stability jumps to perhaps nine days. On day twelve it returns; you recall it again, and stability reaches a month.

Each step multiplies rather than adds. That is why the total number of reviews grows so slowly: the gaps between them grow geometrically while the effort per review stays the same.

If you fail a review, stability drops back, though not all the way. The memory was there; it needs a shorter interval for a while before it can stretch again. Schedulers call this a lapse.

What this sample is for

This document was written for Odyssey's screenshots. It is released with the project under the Apache License 2.0, so you are free to use it to try the reader, make notes, and run a review.